

Introduction - Stable Matching

Lecture 1 - ADA Winter-2023

Diptapriyo Majumdar

January 16, 2022

Outline

- ➊ General Information
- ➋ Algorithm: What is it and why?
- ➌ Wand Selection
- ➍ Course Outline and References

General Information

- **Lectures:** Monday and Thursday
Time Slot: 3 pm - 4:30 pm.
- **Tutorials:** Tuesday (all groups) 1:30 pm - 3 pm.

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No Tutorial tomorrow.



Instructors

- **Lectures:** Diptapriyo Majumdar

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- **Lectures:** Diptapriyo Majumdar
- **Office Hours:** By appointment, Wed 1:30 pm - 2:30 pm (meeting link to be shared)

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- **Office Hours:** By appointment, Wed 1:30 pm - 2:30 pm (meeting link to be shared)
- **Email:** diptapriyo@...

Instructor Team

- **Teaching Fellow (TF):** Riti Choudhury (riti@...)

Instructor Team

- **Teaching Fellow (TF):** Riti Choudhury (riti@...)
- **Teaching Assistant (TA):** to be announced.

Instructor Team

- **Teaching Fellow (TF):** Riti Choudhury (riti@...)
- **Teaching Assistant (TA):** to be announced.
- Each of you will be assigned to one TA who will be clearing your doubts.

How to address query

Your Friend $\rightarrow$ Your TA $\rightarrow$ some other TA $\rightarrow$ Riti $\rightarrow$ me.

Platforms

- **Course Webpage:** Lecture slides, lecture recordings, office hours, any other resource.
- **Google Classroom:** ag6okka
- All announcements will be posted in Google Classroom.
- All evaluated components are to be submitted and graded in this Google Classroom.

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- **Google Classroom:** [ag6okka](#)
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- All evaluated components are to be submitted and graded in this Google Classroom.
- Do not make your solutions public.

Evaluation

- Quiz: 15% (best n-1 policy)

- Homework Assignments (Theory): 15% [Only PDF submission is allowed. Latex template will be shared soon. No zip file will be accepted.]

Latex
Template
will be
shared

- Programming Assignments: 15% (platform to be announced, programming language should be C/C++/Java, not Python) [individual submission]

- Mid-Sem: 25%

- End-Sem: 30%

3 quizzes

3 Theory Assignments

2 Programming Assignments

Evaluation

- **Quizzes:** Short duration and MCQ Quiz.
- **Mid-Sem and End-Sem:**
 - Proctored Exam in offline mode.

Anti-Plagiarism Policy

- **Homeworks and Coding Assignments:**
 - **First instance:** One grade reduction, and 0 in that particular assignment.
 - **Second Instance:** Fail (*F*-grade).
 - **Theory Homeworks:** You can discuss with your friends and/or seniors. If you do, then mention their names. If you find the solutions from the internet, please mention the link where you have found it.
- **Quiz/Exam:** Fail (*F*-grade).

Programming Assignment: Competitive
programming platform [foobars
codeforces]

Late submission policy

- **Theory Assignment:** (7 days time to complete)
 - **1-Day Late submission:** 50% marks reduction in that particular assignment.
 - **≥ 2 days late submission:** 0 in that assignment.
- **Programming Assignment:** No late submission allowed. (2 days to complete)

Other decorum

- Be polite in your emails. We are all working in stressful conditions, and we understand your concerns. Check webpage for details.
- No fooling around or spamming in online platforms.

in google classroom

Outline

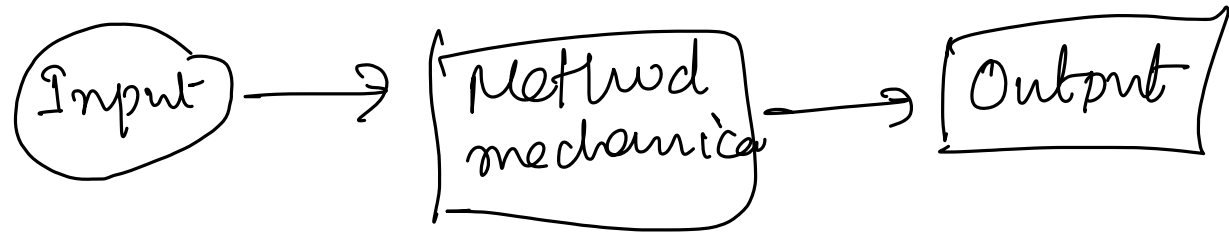
- ① General Information
- ② Algorithm: What is it and why?
- ③ Wand Selection
- ④ Course Outline and References

- **Origin of the word:** Al Khwarizmi.
- Developed mechanical, correct, unambiguous and efficient method for adding, subtracting, multiplying and dividing two integers *A* and *B*.

$$\begin{array}{r}
 \textcircled{1} \\
 5379 \\
 4123 \\
 \hline
 2
 \end{array}$$

- **Origin of the word:** Al Khwarizmi.
- Developed mechanical, correct, unambiguous and efficient method for adding, subtracting, multiplying and dividing two integers A and B .
- Developing mechanical, correct, unambiguous and efficient method to solve problems.
- What have you seen before?

DSA $\rightarrow$ Sorting Algorithms, Spanning Tree



Prerequisites: DSA, DM/DS

Proof by induction, Proof by contradiction
Recurrences

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Wand Selection

- **Input:** n wizards, and n wands. Each wizard lists wands in order of preference. Each wand lists wizards in order of preference.

- **Output:** A stable matching of wizards to wands.

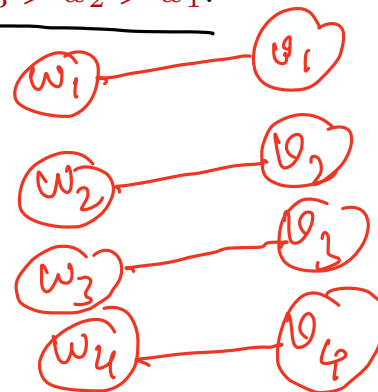
One wizard is allocated exactly one wand.
(and vice versa)

- **Example:** Wizards are w_1, w_2, w_3, w_4 and Wands are v_1, v_2, v_3, v_4 .

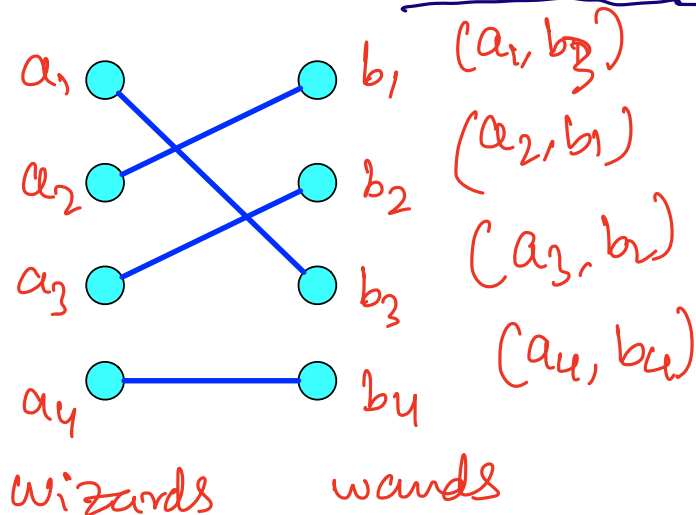
- **Preferences:**

- w_1 prefers $v_1 > v_2 > v_3 > v_4$; w_2 prefers $v_4 > v_3 > v_1 > v_2$; w_3 prefers $v_1 > v_3 > v_2 > v_4$; w_4 prefers $v_4 > v_3 > v_2 > v_1$.
- v_1 prefers $w_1 > w_2 > w_3 > w_4$; v_2 prefers $w_2 > w_4 > w_1 > w_3$; v_3 prefers $w_1 > w_2 > w_4 > w_3$; v_4 prefers $w_4 > w_3 > w_2 > w_1$.

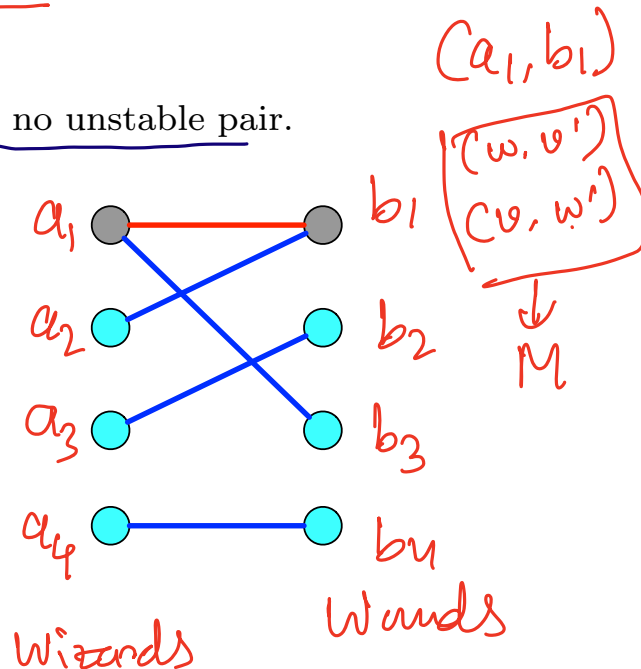
$\{w_1, w_2, w_3, w_4\} \rightarrow$ wizards
 $\{v_1, v_2, v_3, v_4\} \rightarrow$ wands



- **Perfect Matching:** Each wizard is assigned to exactly one wand. Each wand is assigned to exactly one wizard. M
- **Unstable Pair:** Given a matching M , a wizard w and a wand v , i.e. the pair (w, v) is an unstable pair if they prefer each other more than their currently allocated partners.
- **Stable Matching:** A perfect matching with no unstable pair.



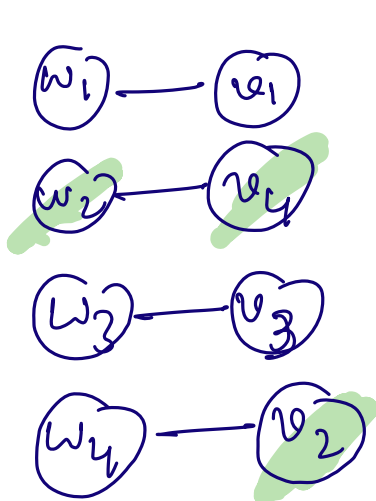
M a_1 prefers b_1 more than b_3
and
 b_1 prefers a_1 more than a_2



Is $(w_1, v_1), (w_2, v_4), (w_3, v_3), (w_4, v_2)$ a stable matching?

No

- w_1 prefers $v_1 > v_2 > v_3 > v_4$; w_2 prefers $v_4 > v_3 > v_1 > v_2$; w_3 prefers $v_1 > v_3 > v_2 > v_4$; w_4 prefers $v_4 > v_3 > v_2 > v_1$.
- v_1 prefers $w_1 > w_2 > w_3 > w_4$; v_2 prefers $w_2 > w_4 > w_1 > w_3$; v_3 prefers $w_1 > w_2 > w_4 > w_3$; v_4 prefers $w_4 > w_3 > w_2 > w_1$.



Is $(w_1, v_4), (v_3, w_1)$

(w_3, v_1)

(w_4, v_4)

(w_4, v_4) is an unstable pair.

(w_4, v_3)

w_4 prefers v_4 more than v_2
 and v_4 prefers w_4 more than w_2

Input: n wizards $w_1, \dots, w_n$ and n wands $v_1, \dots, v_n$.

Initialize all wizards and wands are unmatched;

while some wizard w does not have a wand and w has not tried some wand **do**

$v \leftarrow$ the most preferred wand of w and w has not tried v ;

if v is not holding to any wizard **then**

 Assign v to w (tentative);

end

else

if If v prefers w more than it's current holder w' **then**

 Assign v to w and hence w' becomes free;

end

else

v rejects w ;

end

end

end

(v, w')

} $O(n)$ time

Output the wizard, wand pairs

Example

- w_1 prefers $v_1 > v_2 > v_3 > v_4$; w_2 prefers $v_4 > v_3 > v_1 > v_2$; w_3 prefers $v_1 > v_3 > v_2 > v_4$; w_4 prefers $v_4 > v_3 > v_2 > v_1$.
- v_1 prefers $w_1 > w_2 > w_3 > w_4$; v_2 prefers $w_2 > w_4 > w_1 > w_3$; v_3 prefers $w_1 > w_2 > w_4 > w_3$; v_4 prefers $w_4 > w_3 > w_2 > w_1$.

w_1 ————— v_1

w_2 ~~————— v_2~~
~~————— v_3~~

w_4 ————— v_4

w_4 asks for v_4
 $(w_2, v_4) \rightarrow$ Not unstable
 $(w_2, v_2) \rightarrow$ Not unstable
 (w_4, v_2)
 (w_3, v_3)

- **Observation 1:** Each wizard tries wands at decreasing order of preference.
- **Observation 2:** Each wizard tries each wand at most once.
- **Observation 3:** Once a wand v is assigned to some wizard, then v never becomes free.
- **Question 1:** Does this algorithm terminate?
- **Question 2:** How many steps (in terms of n when n wizards and n wands) does this algorithm take?

while (outer) loop runs at most n times
for every wizard.

There are n wizards.
Outer while loop runs n^2 times.

Hence, algorithm will terminate

→ $O(n^2)$ -

Fact 1: The algorithm terminates with a perfect matching.

Suppose that there is a wizard w^* that is not allocated to any wand after the algorithm terminates. Let v^* is the least preferred preferred wand of w^* . But v^* was assigned to w' s.t. w' is more preferred by v^* than w^* .

Then all wands are allocated to some wizard, but some wizard is not allocated to some wand.

↓
some w_i

Contradiction wand is allocated to two wizards

Fact 2: The algorithm produces a stable matching.

correctness

Suppose not. Then (w^*, v^*) is an unstable
pair and the algorithm outputted M .

then $\left\{ \begin{array}{l} w^* \text{ is assigned to } \hat{v} \\ v^* \text{ is assigned to } \hat{w} \end{array} \right\}$ in M

w^* prefers v^* more than $\hat{v}$

v^* prefers w^* more than $\hat{w}$

v^* was allocated to w^*

But v^* is allocated to $\hat{w}$ to — why?
~~re~~ $\hat{w}$ has tried ~~to~~ v^* and
 v^* prefers $\hat{w}$ more than w^*

Initialize all wizards and wands are unmatched;

while *some wizard w does not have a wand and w has not tried some wand* **do**

$v \leftarrow$ *the most preferred wand of w and w has not tried v ;*

if v *is not holding to any wizard* **then**

 | *Assign v to w (tentative);*

end

else

if *If v prefers w more than it's current holder w'* **then**

 | *Assign v to w and hence w' becomes free;*

end

else

 | *v rejects w ; w remains free.*

end

end

end

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- Introduction. *and Recurrences.*
- Divide and Conquer
- Dynamic Programming
- Greedy Methods
- Graph Algorithms - applications of directed DFS, shortest path, spanning tree.
- Network Flow
- NP-Completeness and Computational Intractability.

References

- Algorithm Design by Kleinberg and Tardos - Chapter 1.

References

→ today's lecture

- Algorithm Design by Kleinberg and Tardos - Chapter 1.
- Introduction to Algorithms by Cormen, Liersen, Rivest, Stein.
- Algorithms by Papadimitrou and Dasgupta.
- Fundamentals of Computer Algorithms by Ellis and Sahni.
- Lecture notes on Algorithms by Jeff Erickson.